

Student Number: _____

St George Girls High School

Trial Higher School Certificate Examination

2017



Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Section I	/10
Section II	
Question 11	/15
Question 12	/15
Question 13	/15
Question 14	/15
Question 15	/15
Question 16	/15
Total	/100

Total Marks – 100

Section I

Pages 2 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided at the back of this paper

Section II

Pages 7 – 12

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section
- Begin each question in a new writing booklet

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

Section I

1. What is the value of $\frac{10}{i|z|}$, if $z = -1 + i$?
 - (A) $-5i\sqrt{2}$
 - (B) $2 - 5i$
 - (C) $5i\sqrt{2}$
 - (D) $2 + 5i$

2. Which of the following are the coordinates of the foci of $9x^2 - 36y^2 = 324$?
 - (A) $(\pm\sqrt{5}, 0)$
 - (B) $(0, \pm\sqrt{5})$
 - (C) $(\pm 3\sqrt{5}, 0)$
 - (D) $(0, \pm 3\sqrt{5})$

3. Consider the region bounded by the y -axis, the line $y = 4$ and the curve $y = x^2$.
If this region is rotated about the line $y = 4$, which expression gives the volume of the solid of revolution?
 - (A) $V = \pi \int_0^4 x^2 dy$
 - (B) $V = 2\pi \int_0^2 (4 - y)x dy$
 - (C) $V = \pi \int_0^2 (4 - y)^2 dx$
 - (D) $V = \pi \int_0^4 (4 - y)^2 dx$

Section I (cont'd)

4. $\int_0^2 \frac{x^2}{\sqrt{x^3+1}} dx.$

(A) $\frac{1}{9}$

(B) $\frac{1}{3}$

(C) $\frac{4}{3}$

(D) 9

5. For the hyperbola $(y+1)^2 - x^2 = 1$, find an expression for $\frac{d^2y}{dx^2}$.

(A) $\frac{x}{(y+1)^3}$

(B) $\frac{1}{(y+1)^3}$

(C) $\frac{x}{y+1}$

(D) $\frac{1}{y+1}$

6. The polynomial $P(x) = 4x^3 + 16x^2 + 11x - 10$ has roots α, β and $\alpha + \beta$. What is the value of $\alpha\beta$?

(A) $\frac{5}{2}$

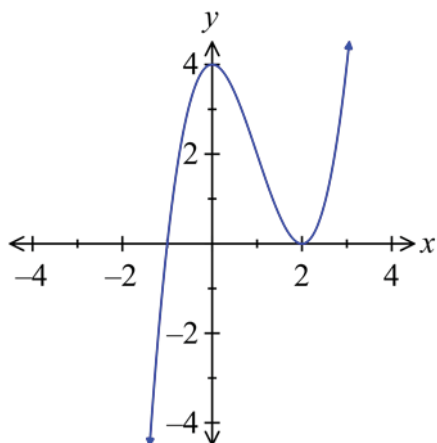
(B) $-\frac{5}{2}$

(C) $\frac{5}{4}$

(D) $-\frac{5}{4}$

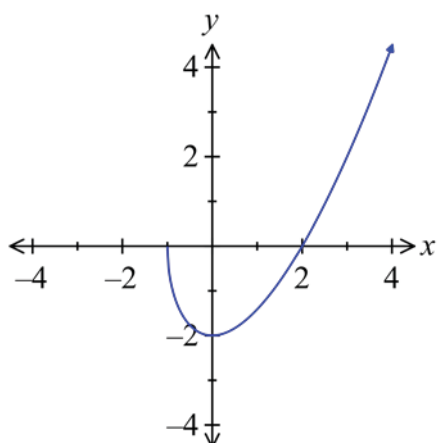
Section I (cont'd)

7. The diagram below shows the graph of the function $y = f(x)$

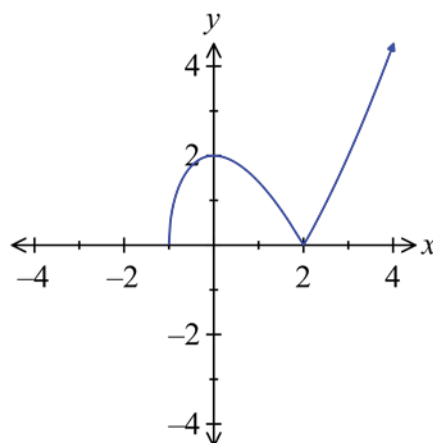


Which of the following is the graph of $y = \frac{1}{f(x)}$?

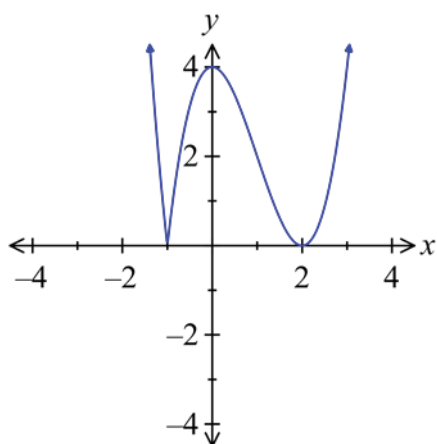
(A)



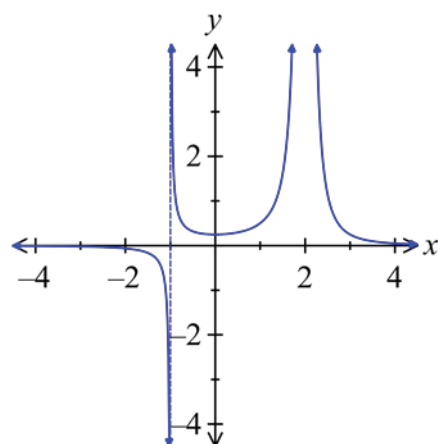
(B)



(C)

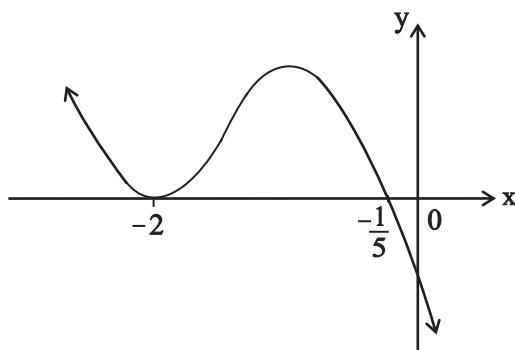


(D)



Section I (cont'd)

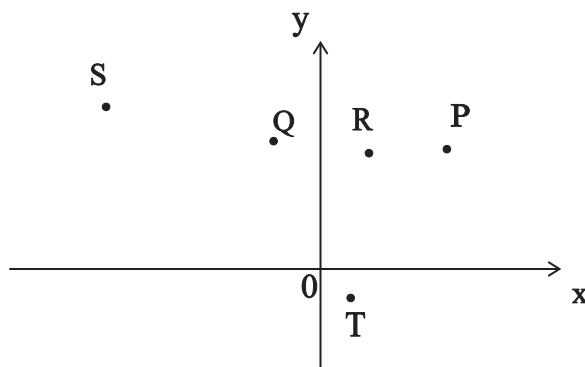
8. The diagram shows the graph of $y = P''(x)$ which is the second derivative of a polynomial $P(x)$.



Which of the following expressions could be $P(x)$?

- (A) $(x + 2)^2 (x - 1)$
- (B) $(x + 2)^4 (x - 1)$
- (C) $(x - 2)^4 (x - 1)$
- (D) $(x - 2)^4 (x + 1)$

9. In the Argand diagram the point **P** represents the complex number z . When this number is divided by $5i$ it gives a new complex number.



Which one of the points on the diagram above represents the new complex number?

- (A) Q
- (B) R
- (C) S
- (D) T
10. The sides of a triangle are the first three terms of an arithmetic progression with the first term 1 and the common difference d . What is the largest set of possible values of d .
- (A) $-1 < d < 1$
- (B) $-\frac{1}{2} < d < 1$
- (C) $-\frac{1}{3} < d < 1$
- (D) $-\frac{1}{4} < d < 1$

End of Section

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet		Marks
(a)	For the complex number $z = \sqrt{2} + \sqrt{2}i$	
(i)	Express z in modulus-argument form	2
(ii)	Find z^{12} .	2
(b)	Evaluate $\int_{\frac{1}{2}}^2 \frac{1}{2x^2 - 2x + 1} dx$ to 4 significant figures.	4
(c)	Find the square root of $1 + 2\sqrt{2}i$.	3
(d)	Reduce the polynomial $x^6 - 9x^3 + 8$ to irreducible factors over the:	
	(i) real field	2
	(ii) complex field	2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Use the substitution $t = \tan \frac{\theta}{2}$ to find $\int \frac{1}{1+\cos \theta+\sin \theta} d\theta$. 3
- (b) $P(a \sec \theta, \tan \theta)$ is a point on the hyperbola $\frac{x^2}{a^2} - y^2 = 1$, $a > 1$, with eccentricity e and asymptotes L_1 and L_2 . M and N are the feet of the perpendiculars from P to L_1 and L_2 respectively. Show that $PM \cdot PN = \frac{1}{e^2}$. 3
- (c) Use integration by parts to find $\int x \sec^2 x dx$. 3
- (d) The area between the curve $y = 3x - x^2$ and $y = x$, between $x = 1$ and $x = 2$ is rotated about the y – axis. Using the method of cylindrical shells, find the volume of the solid of revolution formed. 3
- (e) (i) Given that $\sin x$ can be written as $\sin(2x - x)$ show that
 $\sin x + \sin 3x = 2 \sin 2x \cos x$. 1
(ii) Hence or otherwise find the general solutions of $\sin x + \sin 2x + \sin 3x = 0$. 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

Marks

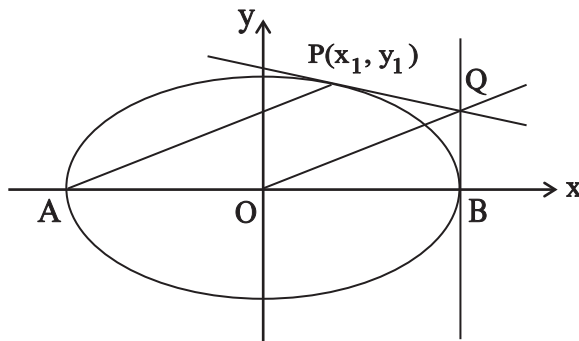
- (a) The roots of $x^3 + x^2 + 1 = 0$ are α, β and γ . Find the cubic equation whose roots are 4

$$\frac{1}{1-\alpha}, \frac{1}{1-\beta}, \frac{1}{1-\gamma}$$

Express your answer in the form $ax^3 + bx^2 + cx + d = 0$.

- (b) (i) On the same Argand diagram carefully sketch the region where
 $|z - 1| \leq |z - 3|$ and $|z - 2| \leq 1$ hold simultaneously. 3
- (ii) Find the greatest possible value for $|z|$ and $\arg z$. 2

- (c) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intersect the x -axis at the points A and B.
 The point P (x_1, y_1) lies on the ellipse. The tangent at P intersects the vertical line passing through B at the point Q as shown in the diagram.



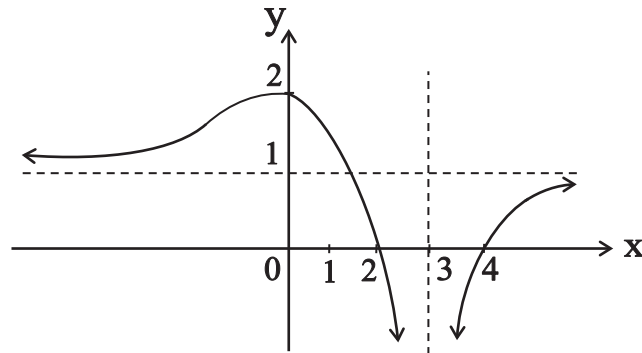
- (i) Show that the equation of tangent at P is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ 2
- (ii) Show that the coordinates of Q are $\left(a, \frac{b^2}{y_1} \left(1 - \frac{x_1}{a}\right)\right)$ 1
- (iii) Show that AP is parallel to OQ 3

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) A sketch of the function $f(x)$ is shown below.



Draw a separate half-page graph for each of the following functions, showing all asymptotes and intercepts.

(i) $y = |f(x)|$ 2

(ii) $y = [f(x)]^2$ 2

(iii) $y = \ln f(x)$ 2

(b) The point $P\left(ct, \frac{c}{t}\right)$ lies on the rectangular hyperbola $xy = c^2$.

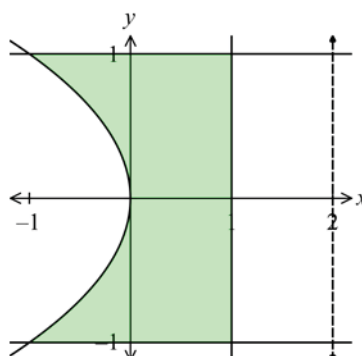
(i) Find the equation of the tangent to the hyperbola at the point P . 2

(ii) The tangent at P cuts the x -axis at A and the y -axis at B . Show that the area of the triangle AOB is independent of t . 2

(c) Find the values of the real numbers p and q given that

$$x^3 + 2x^2 - 15x - 36 = (x + p)^2(x + q)$$
 2

(d) The region shown below is bounded by the lines $x=1$, $y=1$, $y=-1$ and the curve $x=-y^2$. The region is rotated through 360° about the line $x=2$ to form a solid. Calculate the volume of the solid using the method of slicing? 3



End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) A particular solid has as its base the region bounded by the hyperbola $\frac{x^2}{4} - \frac{y^2}{5} = 1$ and the line $x = 4$. 4

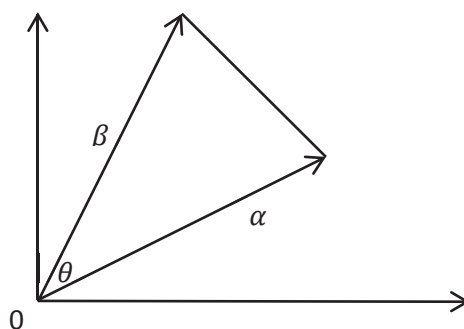
Cross-sections perpendicular to this base and the x –axis are equilateral triangles.
 Find the volume of this solid.

- (b) Let $I_n = \int_1^4 (\sqrt{x} - 1)^n dx$, where $n = 0, 1, 2$.

(i) Show that $(n + 2)I_n = 8 - nI_{n-1}$. 3

(ii) Evaluate I_4 . 3

- (c) (i) Use the results $z + \bar{z} = 2\text{Re}(z)$ and $|z|^2 = z\bar{z}$ for the complex numbers z to show that $|\alpha|^2 + |\beta|^2 - |\alpha - \beta|^2 = 2\text{Re}(\alpha\bar{\beta})$. 3



- (ii) The diagram shows the angle θ between the complex numbers α and β .
 Prove that 2

$$|\alpha||\beta| \cos \theta = \text{Re}(\alpha\bar{\beta}).$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Let $F(x) = e^{x^2}$ for all $x \geq 0$.

(i) Find $F^{-1}(x)$, the inverse function of $F(x)$ 1

(ii) State its domain and range of $F^{-1}(x)$. 2

(iii) On the same set of axes, sketch $F^{-1}(x)$ and $F(x)$ indicate the region represented by 2

$$\int_0^1 F(x)dx \quad \text{and} \quad \int_1^e F^{-1}(x)dx$$

(iv) Evaluate $\int_0^1 F(x)dx + \int_1^e F^{-1}(x)dx$. 2

(b) The cubic equation $x^3 + kx + 1 = 0$, where k is a constant, has roots α, β and γ . For each positive integer n ,

$$S_n = \alpha^n + \beta^n + \gamma^n.$$

(i) State the value of S_1 . 1

(ii) Express S_2 in terms of k . 2

(iii) Show that for all values of n ,

$$S_{n+3} + kS_{n+1} + S_n = 0. \quad 3$$

(iv) Hence or otherwise express $\alpha^5 + \beta^5 + \gamma^5$ in terms of k . 2

End of Examination

Solutions

$$\begin{aligned}
 1. \quad \frac{10}{i/3} &= \frac{10}{i\sqrt{1+i}} \\
 &= \frac{10}{i\sqrt{2}} \times \frac{i\sqrt{2}}{i\sqrt{2}} \\
 &= \frac{10\sqrt{2}i}{-2} \\
 &= -5\sqrt{2}i
 \end{aligned}$$

A

$$2. \quad 9x^2 - 36y^2 = 324$$

$$\frac{x^2}{36} - \frac{y^2}{9} = 1$$

$$b^2 = a^2(e^2 - 1)$$

$$\frac{9}{36} = e^2 - 1$$

$$e^2 = \frac{5}{4}$$

$$e = \frac{\sqrt{5}}{2} \quad a = 6$$

$$\begin{aligned}
 \text{Foci } (\pm ae, 0) &= (\pm 6 \times \frac{\sqrt{5}}{2}, 0) \\
 &= (\pm 3\sqrt{5}, 0)
 \end{aligned}$$

C

3. Using Disc Method

- C

$$4. \quad \int_0^2 \frac{x^2}{\sqrt{x^3+1}} dx = \frac{1}{3} \int_1^9 \frac{3x^2}{\sqrt{x^3+1}} dx$$

$$= \frac{1}{3} \int_1^9 \frac{du}{\sqrt{u}} \quad \begin{array}{l} u = x^3 + 1 \\ \frac{du}{dx} = 3x^2 \end{array}$$

$$= \frac{1}{3} \cdot 2 \left[u^{1/2} \right]_1^9 \quad \begin{array}{l} x=2, u=9 \\ x=0, u=1 \end{array}$$

$$= \frac{4}{3}$$

C

$$5. (y+1)^2 - x^2 = 1$$

$$2(y+1) \frac{dy}{dx} - 2x = 0$$

$$\frac{dy}{dx} = \frac{x}{y+1}$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{y+1 - x \cdot y'}{(y+1)^2} \\ &= \frac{y+1 - \frac{x^2}{y+1}}{(y+1)^2} \end{aligned}$$

$$= \frac{(y+1)^2 - x^2}{(y+1)^3}$$

$$= \frac{1}{(y+1)^3}$$

B

$$6. P(x) = 4x^3 + 16x^2 + 11x - 10$$

$$\Sigma \alpha: \alpha + \beta + \alpha + \beta = \frac{-16}{4}$$

$$= -4$$

$$2(\alpha + \beta) = -4$$

$$\alpha + \beta = -2$$

$$\Sigma \alpha\beta: \alpha\beta + \alpha(\alpha + \beta) + \beta(\alpha + \beta) = \frac{11}{4}$$

$$\alpha\beta + (\alpha + \beta)^2 = \frac{11}{4}$$

$$\alpha\beta + (-2)^2 = \frac{11}{4}$$

$$\begin{aligned} \alpha\beta &= \frac{11}{4} - 4 \\ &= \frac{3}{4} \end{aligned}$$

D

7. D

8. Second derivative has double root at $x=2$.
So the original function should have a
root of multiplicity 4 at $x=-2$

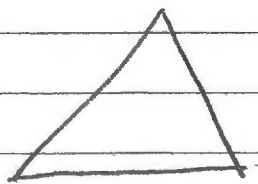
B

9. A division of $5i$ is equivalent to
a multiplication of $\frac{1}{5i} \times \frac{5i}{i} = \frac{i}{-5}$

Multiplying P by $\frac{i}{-5}$ means

D

10.



Sides of triangle
 $1, 1+d, 1+2d$

$$\begin{aligned}\text{Now } 1+d+1 &< 1+2d \\ 2+d &< 1+2d \\ 1 &< d\end{aligned}$$

or

$$\begin{aligned}1+1+2d &< 1+d \\ 2+2d &< 1+d \\ d &< -1\end{aligned}$$

or

$$\begin{aligned}1+d+1+2d &< 1 \\ 2+3d &< 1 \\ 3d &< -1 \\ d &< -\frac{1}{3}\end{aligned}$$

$$\therefore -\frac{1}{3} < d < 1$$

C

MATHEMATICS EXTENSION 2 – QUESTION 11

SUGGESTED SOLUTIONS

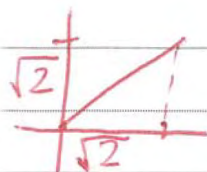
MARKS

MARKER'S COMMENTS

a) i $|z| = \sqrt{2+2}$
 $= 2$

$\arg z = \frac{\pi}{4}$

$\therefore z = 2 \operatorname{cis} \frac{\pi}{4}$



1

1

ii $z^{12} = 2^{12} \operatorname{cis} \left(\frac{\pi}{4} \times 12 \right)$

1, 1

$= 4096 (\cos 3\pi + i \sin 3\pi)$

$= 4096 (-1 + 0i)$

$= -4096$

b) $\int_{\frac{1}{2}}^2 \frac{1}{2x^2 - 2x + 1} dx = \frac{1}{2} \int_{\frac{1}{2}}^2 \frac{1}{x^2 - x + \frac{1}{2}} dx$

$= \frac{1}{2} \int_{\frac{1}{2}}^2 \frac{1}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} dx$

1

$= \frac{1}{2} \left[\frac{1}{\frac{1}{2}} \tan^{-1} \frac{\left(x - \frac{1}{2}\right)}{\frac{1}{2}} \right]_{\frac{1}{2}}^2$

1

$= \frac{1}{2} \left[2 \tan^{-1} (2x - 1) \right]_{\frac{1}{2}}^2$

$= \tan^{-1} 3 - \tan^{-1} 0$

1

$= 1.249045 \dots$

≈ 1.249

1

Note that trigonometric calculus is performed in radians.

c) Let $x+iy$ be the square root of $1+2\sqrt{2}i$

The equivalent in degrees (71.57°) was awarded $3\frac{1}{2}$

$\therefore (x+iy)^2 = 1+2\sqrt{2}i \quad x, y \in \mathbb{R}$

$x^2 - y^2 + 2ixy = 1 + 2\sqrt{2}i$

Equating coefficients,

$x^2 - y^2 = 1$

$2xy = 2\sqrt{2}$

1

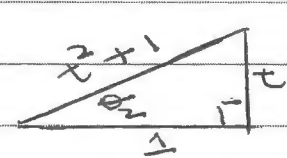
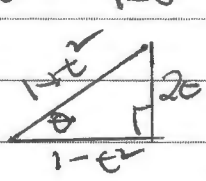
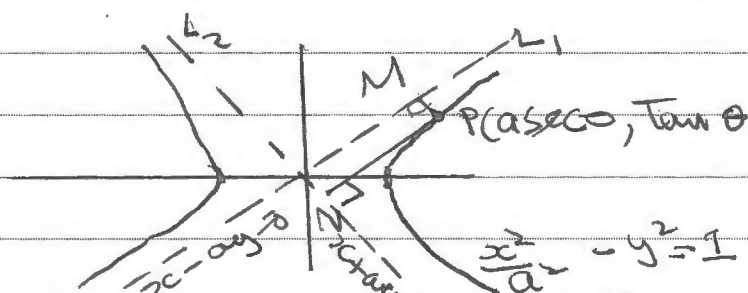
$xy = \sqrt{2}$

$y = \frac{\sqrt{2}}{x}$

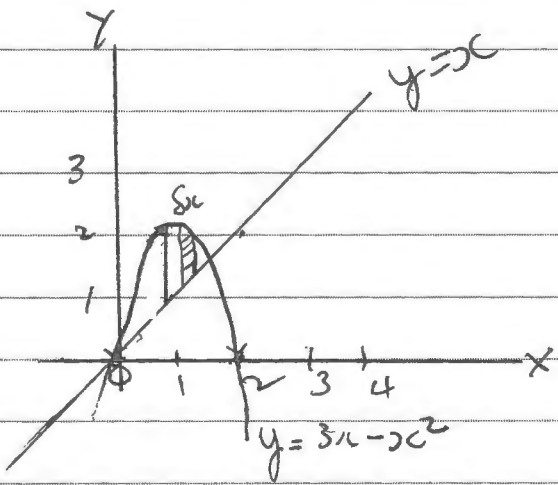
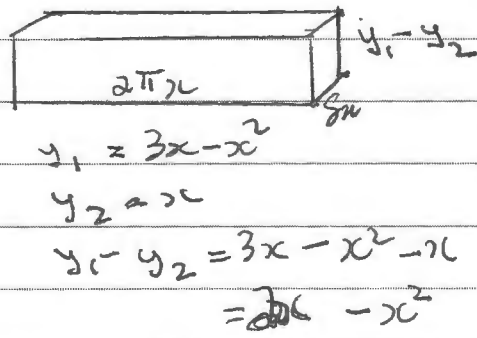
MATHEMATICS EXTENSION 2 – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
sub $y = \frac{\sqrt{2}}{x}$ into $x^2 - y^2 = 1$:		
$x^2 - \frac{2}{x^2} = 1$		Note that $\pm\sqrt{2} \pm i \neq \pm(\sqrt{2} + i)$, as the LHS suggests 4 solutions.
$x^4 - 2 = x^2$		
$x^4 - x^2 - 2 = 0$		
$(x^2 - 2)(x^2 + 1) = 0$		
$(x - \sqrt{2})(x + \sqrt{2})(x^2 + 1) = 0$		
$\therefore x = \pm\sqrt{2} \quad (\because x \in \mathbb{R})$	1	
$\therefore y = \pm 1$		
$\therefore$ the square root of $1 + 2\sqrt{2}i$ is $\pm(\sqrt{2} + i)$	1	
i.e. $\sqrt{2} + i$ and $-\sqrt{2} - i$		
d) i $x^6 - 9x^3 + 8 = (x^3)^2 - 9x^3 + 8$		This is a degree 6 polynomial; long division was a bad choice.
$= (x^3 - 1)(x^3 - 8)$		
$= (x - 1)(x^2 + x + 1)(x - 2)(x^2 + 2x + 4)$	1, 1	
irreducible over $\mathbb{R}$		
ii For $x^2 + x + 1$,		
$x = \frac{-1 \pm \sqrt{3}i}{2}$		
$= -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$		
for $x^2 + 2x + 4$,		
$x = \frac{-2 \pm 2\sqrt{3}i}{2}$		
$= -1 \pm \sqrt{3}i$		
$\therefore x^6 - 9x^3 + 8 = (x - 1)\left(x - \left[-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i\right]\right)(x - 2)\left(x - \left[-1 \pm \sqrt{3}i\right]\right)$		
$= (x - 1)\left(x + \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)\left(x + \frac{1}{2} - \frac{\sqrt{3}}{2}i\right)(x - 2)(x + 1 + \sqrt{3}i)(x + 1 - \sqrt{3}i)$	1, 1	
errors of sign were worth $\frac{1}{2}$ a mark.		

MATHEMATICS EXTENSION 2 – QUESTION

12a)	SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
	$t = \tan \frac{\theta}{2}$ $\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$ $\frac{d\theta}{dt} = \frac{2}{t^2 + 1}$ $d\theta = \frac{2 dt}{t^2 + 1}$		
	$\sin^2 \theta + \cos^2 \theta = 1$ $\tan^2 \theta + 1 = \sec^2 \theta$		
		1	
	$\int \frac{1}{1 + \cos \theta + \sin \theta} d\theta$ $= \int \frac{1}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$ $= \int \frac{2 dt}{(1+t^2) + (1-t^2) + 2t}$ $= \int \frac{2 dt}{2 + 2t}$ $= \int \frac{1}{1+t} dt$ $= \ln 1+t + C$ $= \ln 1 + \tan \frac{\theta}{2} + C$	1	
	$\tan \theta = \frac{2t}{1-t^2}$ 		
	$\cos \theta = \frac{1-t^2}{1+t^2}$ $\sin \theta = \frac{2t}{1+t^2}$	1	
12b)			
	Asymptotes: $y = \pm \frac{b}{a}x$ $\therefore y = \pm \frac{x}{a}$ $\frac{x^2}{a^2} - y^2 = 1$ $b^2 = 1$ $b = 1$		
	$L_1: a \cdot y > x$ $\therefore x - ay < 0$		
	$L_2: ay < -x$ $\therefore x + ay < 0$	1	
	Using perpendicular formula for distance of PM and PN		
	$PM = \left \frac{ax + by + c}{\sqrt{a^2 + b^2}} \right $ $= \left \frac{a \cdot \sec \theta - a \cdot \tan \theta}{\sqrt{1 + a^2}} \right $		
	$PN = \left \frac{a \cdot \sec \theta + a \cdot \tan \theta}{\sqrt{1 + a^2}} \right $		
	$PM \times PN = \left \frac{a(\sec \theta - \tan \theta) \times a(\sec \theta + \tan \theta)}{\sqrt{1 + a^2}} \right $ $= \frac{a^2}{\sqrt{1 + a^2}} (\sec^2 \theta - \tan^2 \theta)$		

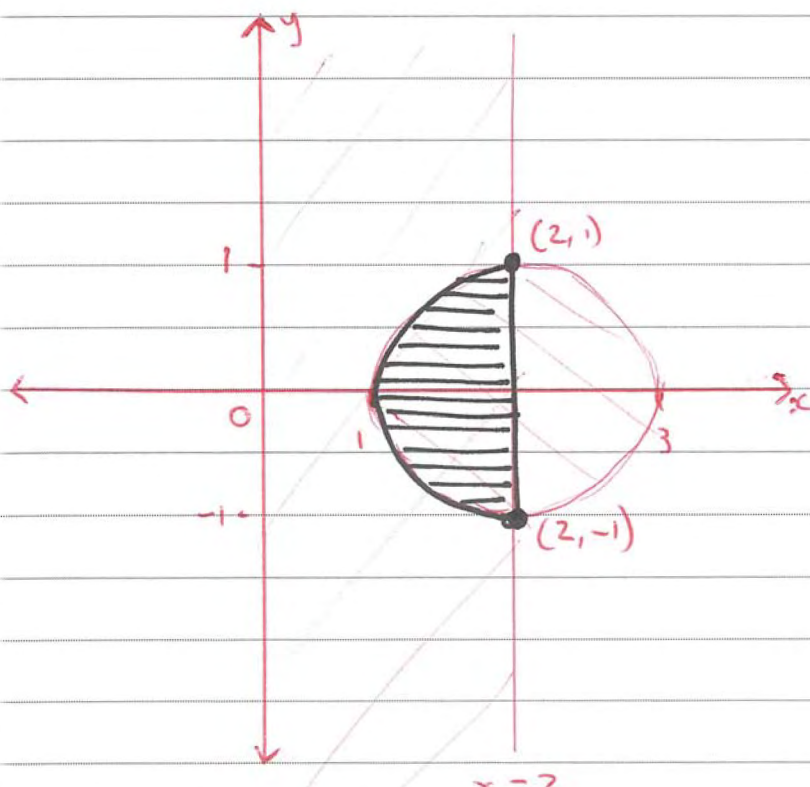
MATHEMATICS EXTENSION 2 – QUESTION

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
12b) cont $PM \times PN = \frac{a^2}{1+a^2} \times 1$ $= \frac{a^2}{1+a^2}$ but $b^2 = a^2(e^2 - 1)$ $b=1, 1 = a^2(e^2 - 1)$ $1 = a^2 e^2 - a^2$ $1 + a^2 = a^2 e^2$ $e^2 = a$ $\therefore PM \times PN = \frac{1}{e^2} \leftarrow$	1 1 1 1 1	
12c) $\int x \cdot \sec^2 x \, dx$ $u=x$ $dv=\sec^2 x$ $= uv - \int v \cdot du$ $du=1$ $v=\tan x$ $= x \cdot \tan x - \int \tan x \, dx$ $= x \cdot \tan x - \int \frac{\sin x}{\cos x} \, dx$ $= x \cdot \tan x + \int \frac{-\sin x}{\cos x} \, dx$ $= x \cdot \tan x + \ln \cos x + c$	1 1 1 1	
12d) $y = 3x - x^2$ $y = x$ $x = 3x - x^2$ $x^2 + x - 3x = 0$ $x^2 - 2x = 0$ $x(x - 2) = 0$ $x = 0$ or 2		1
$A = 2\pi x \cdot \delta x$ $\delta V = 2\pi x y \cdot \delta x$ $= 2\pi x (2x - x^2) \cdot \delta x$ $V = \lim_{\delta x \rightarrow 0} 2\pi x (2x - x^2) \cdot \delta x$ $= \lim_{\delta x \rightarrow 0} 2\pi (2x^2 - x^3)$ $= 2\pi \int (2x^2 - x^3) \, dx$ $= 2\pi \left[\frac{2}{3}x^3 - \frac{x^4}{4} + c \right]$ $= \frac{11\pi}{6}$ cu units	 $y_1 = 3x - x^2$ $y_2 = x$ $y_1 - y_2 = 3x - x^2 - x$ $= 2x - x^2$	1 1

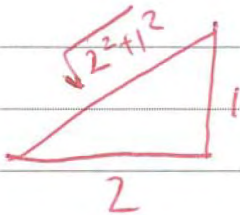
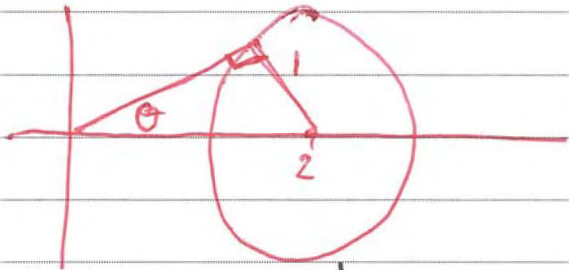
MATHEMATICS EXTENSION 2 – QUESTION

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
12e)i) $\sin x + \sin 3x$ $= \sin(2x-x) + \sin(2x+x)$ $= \sin 2x \cdot \cos x - \cos 2x \cdot \sin x + \sin 2x \cdot \cos x + \cos 2x \cdot \sin x$ $= 2 \sin 2x \cdot \cos x$ $\therefore \sin x + \sin 3x = 2 \sin 2x \cdot \cos x$	1	
ii) $\sin x + \sin 2x + \sin 3x = 0$ $2 \sin 2x \cdot \cos x + \sin 2x = 0$ $\sin 2x (2 \cos x + 1) = 0$ $\sin 2x = 0$ or $2 \cos x + 1 = 0$ $x = n\pi$ or $2 \cos x = -1$ $\cos x = -\frac{1}{2}$ $x = 2n\pi \pm \frac{2\pi}{3}$ OR $x = (2n+1)\pi \pm \frac{\pi}{3}$ $\therefore x = \frac{n\pi}{2}$ OR $x = 2n\pi \pm \frac{2\pi}{3}$ (or $(2n+1)\pi \pm \frac{\pi}{3}$)	1	

MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) Let $x = \frac{1}{1-d}$ $1-d = \frac{1}{x}$ $d = 1 - \frac{1}{x}$ $= \frac{x-1}{x}$ $\therefore$ cubic is $\left(\frac{x-1}{x}\right)^3 + \left(\frac{x-1}{x}\right)^2 + 1 = 0$ $\frac{x^3-3x^2+3x-1}{x^3} + \frac{x^2-2x+1}{x^2} + 1 = 0$ $x^3 - 3x^2 + 3x - 1 + x^3 - 2x^2 + x + x^3 = 0$ $3x^3 - 5x^2 + 4x - 1 = 0$	1 1 1 1	
b) i 	1 1 1	for $x=2$ for $(x-2)^2 + y^2 = 1$ for correct region

MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b) <u>ii</u>  Max z occurs at $(2, 1)$ [or $(2, -1)$] $ z = \sqrt{2^2 + 1^2}$ $= \sqrt{5}$ Max $\arg z$ occurs when tangent is perpendicular to radius  $\sin \theta = \frac{1}{2}$ $\theta = \frac{\pi}{6}$	1	Best responses came from students who redrew the relevant diagrams. Note that if your answer to b) i was incorrect, it was difficult to demonstrate the required skills for part ii Very few students correctly answered this part.
c) <u>i</u> $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{d}{dx} \left[\frac{x^2}{a^2} + \frac{y^2}{b^2} \right] = \frac{d}{dx} (1)$ $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{-2x/a^2}{2y/b^2}$ $= \frac{-x b^2}{y a^2}$ At $P(x_1, y_1)$, $\frac{dy}{dx} = \frac{-x_1 b^2}{y_1 a^2}$	1	correct differentiation

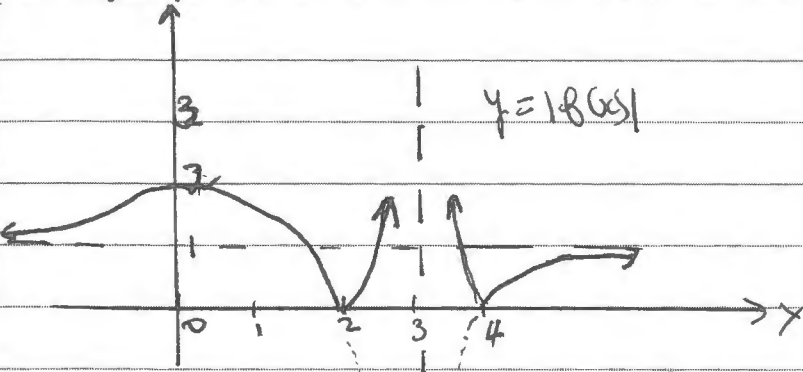
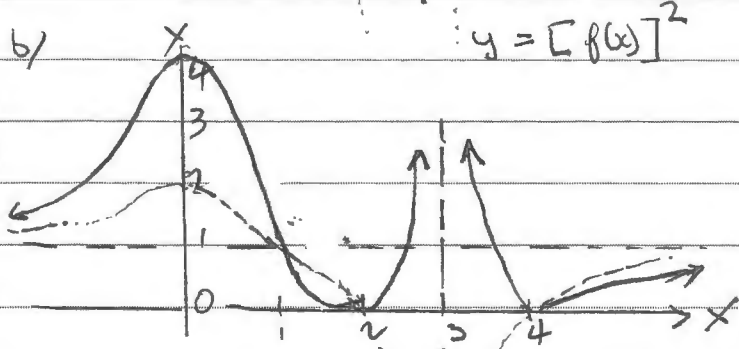
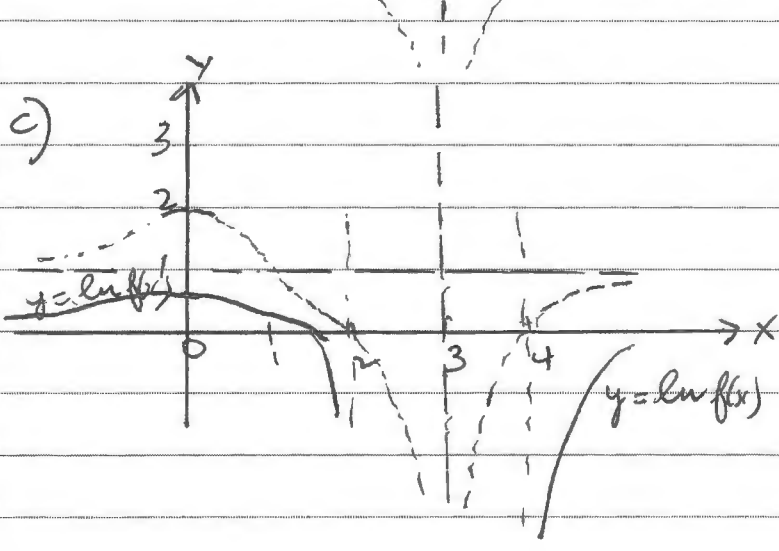
MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
Equation of tangent through (x_1, y_1) :		
$y - y_1 = \frac{-x_1 b^2}{y_1 a^2} (x - x_1)$		
$yy_1 a^2 - y_1^2 a^2 = -xx_1 b^2 + x_1^2 b^2$		
$xx_1 b^2 + yy_1 a^2 = x_1^2 b^2 + y_1^2 a^2$		
$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$		
$= 1$		
[Because $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$, as $P(x_1, y_1)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$]		
$\therefore \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	1	correct substitution and algebra
Q ii The coordinates of B are $(a, 0)$ $\therefore Q$ lies on the line $x = a$		
Sub $x = a$ into equation of tangent		
$\frac{ax_1}{a^2} + \frac{yy_1}{b^2} = 1$		
$\frac{x_1}{a} + \frac{yy_1}{b^2} = 1$		
$\frac{yy_1}{b^2} = 1 - \frac{x_1}{a}$		

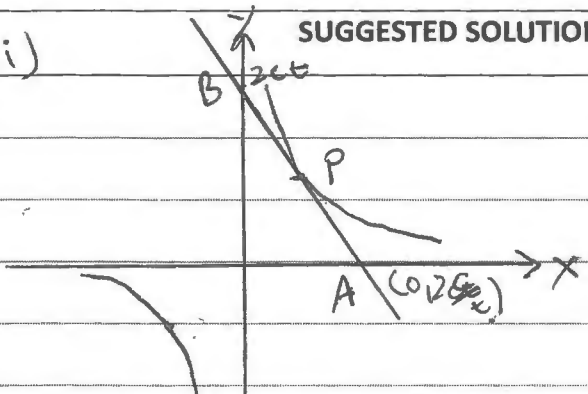
MATHEMATICS EXTENSION 2 – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$y = \frac{b^2}{y_1} \left(1 - \frac{x_1}{a}\right)$ $\therefore Q \left(a, \frac{b^2}{y_1} \left(1 - \frac{x_1}{a}\right)\right)$	1	substitution of $x=a$ and correct algebra.
$\textcircled{ii} \quad m_{AP} = \frac{y_1 - 0}{x_1 - -a}$ $= \frac{y_1}{x_1 + a}$	1	
$m_{OQ} = \frac{\frac{b^2}{y_1} \left(1 - \frac{x_1}{a}\right) - 0}{a - 0}$ $= \frac{b^2}{y_1} \times \frac{a - x_1}{a} \times \frac{1}{a}$ $= \frac{b^2(a - x_1)}{a^2 y_1} \quad \textcircled{1}$	1	simplified expression for m_{OQ}
Since (x_1, y_1) lies on the ellipse, $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$ $b^2 x_1^2 + a^2 y_1^2 = a^2 b^2$ $a^2 y_1^2 = a^2 b^2 - x_1^2 b^2$ $= b^2 (a^2 - x_1^2)$ $= b^2 (a - x_1)(a + x_1)$ $\therefore \frac{a^2 y_1^2}{a + x_1} = b^2 (a - x_1) \quad \textcircled{2}$		
sub $\textcircled{2}$ into $\textcircled{1}$ $m_{OQ} = \frac{a^2 y_1^2}{a^2 y_1 (a + x_1)}$ $= \frac{y_1}{a + x_1} = m_{AP} \quad \therefore AP \parallel OQ$	1	correct algebra

MATHEMATICS EXTENSION 2 – QUESTION

14a)	SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
		2	
b)		2	
c)		2	
d) $P(ct, \frac{c}{e})$ $xy = c$			
i)	$y = c \cdot x^{-1}$ $\frac{dy}{dx} = -c \cdot x^{-2}$ $= -\frac{c}{x^2}$		alternative implicit differentiation
grad at $P(ct, \frac{c}{e}) = -\frac{c}{c^2 e} = -\frac{1}{e^2}$		1	
$y - \frac{c}{e} = -\frac{1}{e^2} (x - ct)$			
$e^2(y - \frac{c}{e}) = -x + ct$			
$e^2 y - ct = -x + ct$			
$x + e^2 y - 2ct = 0$		1	

MATHEMATICS EXTENSION 2 – QUESTION

14) ii)	SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
			
At A, $y=0 \Rightarrow x + t^2 y - 2ct = 0$ $x - 2ct = 0$ $\therefore x = 2ct$ A is $(2ct, 0)$		1	
At B, $x=0 \Rightarrow t^2 y - 2ct = 0$ $y = \frac{2ct}{t^2}$ $y = \frac{2c}{t}$ B is $(0, \frac{2c}{t})$			
Area $\triangle AOB = \frac{1}{2} \times 2ct \times \frac{2c}{t}$ $= 2c^2$ $\therefore$ Area is independent of t .			
14) $x^3 + 2x^2 - 15x - 36 = (x+p)^2 \cdot (x+q)$ - p is a root of multiplicity 2 $f(x) = x^3 + 2x^2 - 15x - 36$ $f'(x) = 3x^2 + 4x - 15$ - p is a root of $f'(x) = 0$ of multiplicity 1 $3x^2 + 4x - 15 = 0$ $(3x - 5)(x + 3) = 0$ $x = \frac{5}{3}$ or $x = -3$			
$f(\frac{5}{3}) = (\frac{5}{3})^3 + 2 \times (\frac{5}{3})^2 - 15 \times \frac{5}{3} - 36 \neq 0$ $f(-3) = (-3)^3 + 2 \times (-3)^2 - 15 \times -3 - 36 = 0$ $\therefore p = -3$		1	
Let $x=0$, $-36 = p^2 \cdot q$ $-36 = 9 \cdot q$ $q = -4$		1	
$\therefore p = -3, q = -4$			

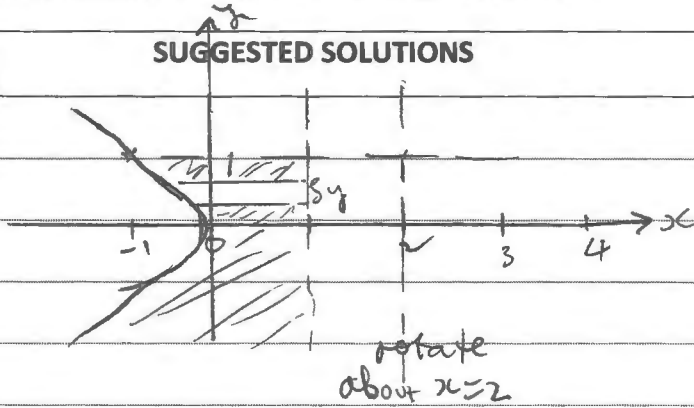
MATHEMATICS EXTENSION 2 – QUESTION

14 (a)

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS



$$r = \text{inner radius} = 1$$

$$R = \text{outer radius} = |x| + 2 = y^2 + 2$$

$$\begin{aligned} \text{Area of slice} &= \pi(R^2 - r^2) \\ &= \pi(y^2 + 2)^2 - 1^2 \end{aligned}$$

$$= \pi(y^4 + 4y^2 + 4 - 1)$$

$$= \pi(y^4 + 4y^2 + 3)$$

1

$$\delta V = \pi(y^4 + 4y^2 + 3) \cdot \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=-1}^1 \pi(y^4 + 4y^2 + 3) \cdot \delta y$$

1

$$= 2\pi \int_0^1 (y^4 + 4y^2 + 3) dy$$

$$= 2\pi \left[\frac{y^5}{5} + \frac{4y^3}{3} + 3y + c \right]_0^1$$

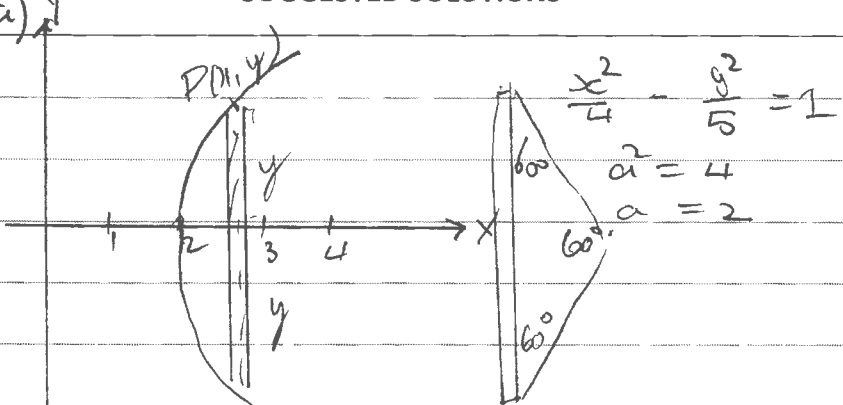
$$= 2\pi \left(\frac{1}{5} + \frac{4}{3} + 3 + c \right) - (0 + c)$$

$$= 2\pi \times \frac{68}{15}$$

$$= \frac{136}{15} \pi \text{ cm units.}$$

1

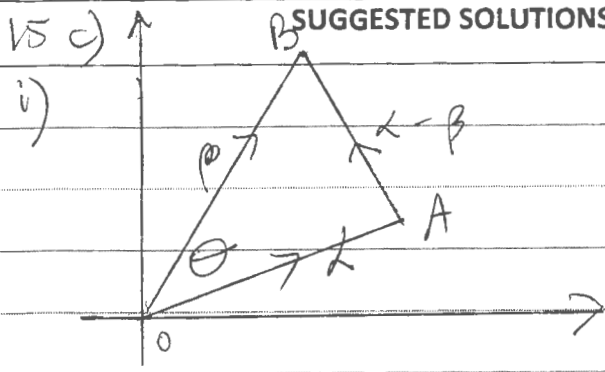
MATHEMATICS EXTENSION 2 – QUESTION

15a)	SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
	 $\frac{x^2}{4} - \frac{y^2}{5} = 1$ $a^2 = 4$ $a = 2$		
	Take $P(x, y)$ on the curve, $y > 0$ Area of cross-section $A(x) = \frac{1}{2} ab \cdot \sin C$ $= \frac{1}{2} \cdot 2y \cdot 2y \cdot \sin 60^\circ$ $= 2y^2 \times \frac{\sqrt{3}}{2}$ $A(x) = \sqrt{3} \cdot y^2$	1	
	$\delta V = \sqrt{3} \cdot y^2 \cdot \delta x$ $V = \lim_{\delta x \rightarrow 0} \sqrt{3} \cdot y^2 \cdot \delta x$	1	
	Now $\frac{x^2}{4} - \frac{y^2}{5} = 1$ $\frac{x^2}{4} + 1 = \frac{y^2}{5}$ $y^2 = 5 \left(\frac{x^2}{4} + 1 \right)$	1	
	$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=2}^4 \sqrt{3} \cdot 5 \left(\frac{x^2}{4} + 1 \right) \delta x$ $= 5\sqrt{3} \int_2^4 \left(\frac{x^2}{4} + 1 \right) dx$ $= 5\sqrt{3} \left[\frac{x^3}{12} - x + c \right]_2^4$ $= 5\sqrt{3} \left[\frac{64}{12} - 4 - \left(\frac{8}{12} - 2 \right) \right]$ $= 5\sqrt{3} \times \frac{8}{3}$ $= \frac{40\sqrt{3}}{3} \text{ units}^3$	1	
	15b) $I_n = \int_1^4 (\sqrt{x} - 1)^n dx, n = 0, 1, 2, \dots$ i) $u = (x^{\frac{1}{2}} - 1)^n$ $dv = 1$ $du = n(x^{\frac{1}{2}} - 1)^{n-1} \times \frac{1}{2} x^{-\frac{1}{2}}$ $= \frac{n(x^{\frac{1}{2}} - 1)^{n-1}}{2\sqrt{x}}$ $I_n = uv - \int v du$	1	

MATHEMATICS EXTENSION 2 – QUESTION

	SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
I_n	$= [x(\sqrt{x}-1)^n + c]_1^4 - \int_1^4 x \cdot \frac{n}{2} \frac{(\sqrt{x}-1)}{\sqrt{x}} dx$ $= [4 \times (2-1) - 0] - \frac{n}{2} \int_1^4 \sqrt{x} (\sqrt{x}-1)^{n-1} dx$ $= 4 - \frac{n}{2} \int_1^4 ((\sqrt{x}-1) + 1)(\sqrt{x}-1)^{n-1} dx$ $= 4 - \frac{n}{2} \int_1^4 ((\sqrt{x}-1)(\sqrt{x}-1)^{n-1} + 1 \times (\sqrt{x}-1)^{n-1}) dx$ $= 4 - \frac{n}{2} \int_1^4 ((\sqrt{x}-1)^n + (\sqrt{x}-1)^{n-1}) dx$ $= 4 - \frac{n}{2} \int_1^4 (\sqrt{x}-1)^n - \frac{n}{2} \int_1^4 (\sqrt{x}-1)^{n-1} dx$	1	
I_n	$I_n = 4 - \frac{n}{2} I_n - \frac{n}{2} I_{n-1}$ $2 \cdot I_n = 8 - n \cdot I_n - n \cdot I_{n-1}$ $\therefore (n+2) \cdot I_n = 8 - n \cdot I_{n-1}$	1	
15 b) ii)	Put $n=2$, $4 I_2 = 8 - 2 \cdot I_1$ $I_2 = 2 - \frac{1}{2} I_1$ $I_2 = 2 - \frac{1}{2} \int_1^4 (x^{\frac{1}{2}} - 1) dx$ $= 2 - \frac{1}{2} \left[\frac{2}{3} x^{\frac{3}{2}} - x + c \right]_1^4$ $= 2 - \frac{1}{2} \left[\left(\frac{2}{3} \times 8 - 4 \right) - \left(\frac{2}{3} - 1 \right) \right]$ $= 2 - \frac{9}{6}$ $= \frac{7}{6}$	1	
	Put $n=3$, $I = 8 - 3 \cdot I_2$ $5 \cdot I_3 = 8 - 3 \times \frac{7}{6}$ $I_3 = \frac{9}{10}$	1	
	Put $n=4$, $6 \cdot I_4 = 8 - 4 \cdot I_3$ $= 8 - 4 \times \frac{9}{10}$ $I_4 = \frac{1}{2} \times \frac{22}{5}$ $\therefore I_4 = \frac{11}{5}$	1	

MATHEMATICS EXTENSION 2 – QUESTION

15 c) $\uparrow$ i) 	SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$z + \bar{z} = 2 \operatorname{Re}(z)$, $ z = z \bar{z}$			
Prove $ z ^2 + \beta ^2 - \alpha - \beta ^2 = 2 \operatorname{Re}(\alpha \bar{\beta})$			
$\text{LHS} = z \cdot \bar{z} + \beta \bar{\beta} - (\alpha - \beta)(\bar{\alpha} - \bar{\beta})$	1		
$= z \cdot \bar{z} + \beta \bar{\beta} - (\alpha \bar{\alpha} - \alpha \bar{\beta} - \beta \bar{\alpha} + \beta \bar{\beta})$			
$= z \cdot \bar{z} + \beta \bar{\beta} - (\alpha \bar{\alpha} - \alpha \bar{\beta} - \beta \bar{\alpha} + \beta \bar{\beta})$			
$= \cancel{z \cdot \bar{z}} + \cancel{\beta \bar{\beta}} - \cancel{\alpha \bar{\alpha}} + \alpha \bar{\beta} + \beta \bar{\alpha} - \cancel{\beta \bar{\beta}}$	1		
$= \alpha \bar{\beta} + \beta \bar{\alpha}$			
$= \alpha \bar{\beta} + \overline{\alpha \bar{\beta}}$	1		
$= 2 \operatorname{Re}(\alpha \bar{\beta})$	1		
ii) $AB = (\alpha - \beta)$			
Using cosine rule			
$\cos \theta = \frac{ z ^2 + \beta ^2 - \alpha - \beta ^2}{2 \cdot z \cdot \beta }$	1		
$\cos \theta = \frac{2 \operatorname{Re}(\alpha \bar{\beta})}{2 \cdot z \cdot \beta }$			
$\cos \theta = \frac{\operatorname{Re}(\alpha \bar{\beta})}{ z \cdot \beta }$			
$ z \beta \cdot \cos \theta = \operatorname{Re}(\alpha \bar{\beta})$	1		

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

a) i $F(x) = e^{x^2}, x \geq 0$

let $y = e^{x^2}$

for inverse, $x = e^{y^2}, y \geq 0$

$\ln x = y^2$

$y = \sqrt{\ln x} \quad (\because y \geq 0)$

1

Errors of sign were worth $\frac{1}{2}$ a mark

ii $\ln x \geq 0$

$\therefore x \geq 1$

$\therefore$ domain: all real $x, x \geq 1$

range: all real $y, y \geq 0$

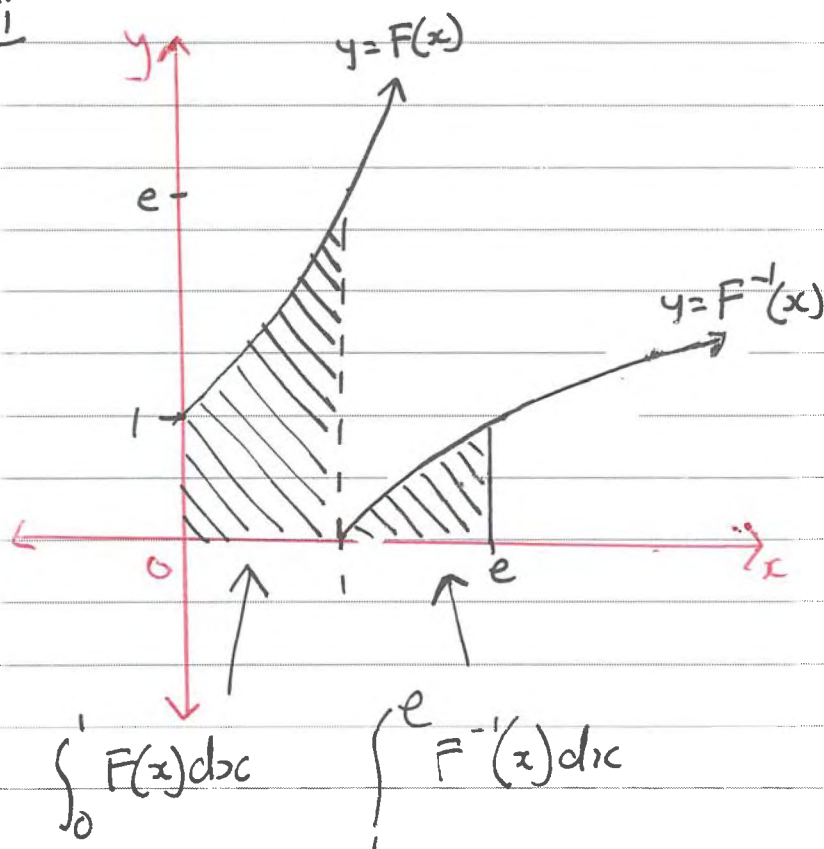
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No half marks were awarded.

1

Note that if you got part i wrong, it was difficult to demonstrate the skills for part ii

iii



1 for curves

1 for regions

(or one correct curve and its region = 1 mark)

iv) Area_C = Area_B by symmetry

$\therefore \int_0^1 F(x) dx + \int_1^e F^{-1}(x) dx = \text{area of rectangle}$

$= e \times 1$

$= e \text{ units}^2$

1

1

MATHEMATICS EXTENSION 2 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b) i $S_1 = \alpha^1 + \beta^1 + r^1$ $= \frac{-b}{a}$ $= \frac{0}{1}$ $= 0$	1	The answer to this question is a <u>value</u> , not an expression.
ii $S_2 = \alpha^2 + \beta^2 + r^2$ $= (\alpha + \beta + r)^2 - 2(\alpha\beta + \alpha r + \beta r)$ $= \frac{-b}{a} - 2(\frac{c}{a})$ $= 0 - 2k$ $= -2k$	1	
iii LHS = $S_{n+3} + kS_{n+1} + S_n$ $= \alpha^{n+3} + \beta^{n+3} + r^{n+3} + k(\alpha^{n+1} + \beta^{n+1} + r^{n+1}) + \alpha^n + \beta^n + r^n$ $= \alpha^{n+3} + k\alpha^{n+1} + \alpha^n + \beta^{n+3} + k\beta^{n+1} + \beta^n + r^{n+3} + kr^{n+1} + r^n$ $= \alpha^n(\alpha^3 + k\alpha + 1) + \beta^n(\beta^3 + k\beta + 1) + r^n(r^3 + kr + 1)$ $= \alpha^n(0) + \beta^n(0) + r^n(0)$ $= 0$ ($\because \alpha, \beta, \text{ and } r \text{ are roots}$) $= \text{RHS}$	1	Note that α being a root does not imply that α^n is also a root.
iv When $n=0$, $S_3 + kS_1 + S_0 = 0$ $S_3 = -kS_1 - S_0$ $= -k(0) - (\alpha^0 + \beta^0 + r^0)$ $= -3$	1	
When $n=2$ $S_5 + kS_3 + S_2 = 0$ $S_5 = -kS_3 - S_2$ $= -k(-3) - (-2k)$ $\therefore S_5 = 5k$ $\therefore \alpha^5 + \beta^5 + r^5 = 5k$	1	